

Small Enterprise Network Simulation

Cisco Packet Tracer — Project Technical Report

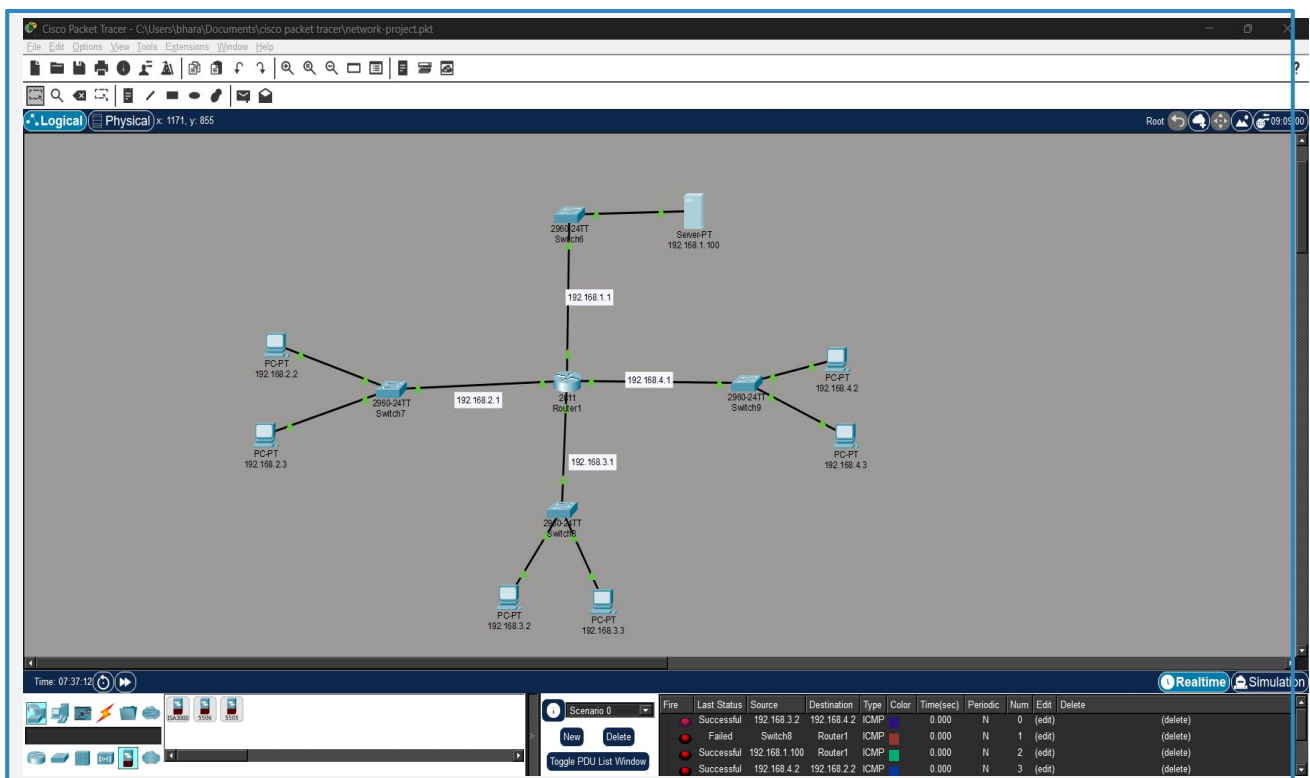


Figure 1 — Network Topology (Cisco Packet Tracer)

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Tool	Cisco Packet Tracer 8.x
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1. PROJECT OVERVIEW

This project involves the design, implementation, and verification of a small enterprise network using Cisco Packet Tracer. The simulation replicates a realistic corporate environment with multiple departments connected through a central router, each residing in its own subnet. A dedicated server provides DNS resolution and HTTPS web services accessible from all departments.

The project demonstrates core networking competencies including IP addressing, subnetting, inter-VLAN routing, DNS configuration, and HTTPS service deployment — skills directly relevant to roles in network administration, SOC operations, and cybersecurity engineering.

Project Name	Small Enterprise Network Simulation
Simulation Tool	Cisco Packet Tracer 8.x
Router Model	Cisco 2811
Switch Model	Cisco 2960-24TT (x4)
End Devices	7 PC-PT workstations + 1 Server-PT
Departments	4 (Management/IT, Dept-2, Dept-3, Dept-4)
Services	DNS, HTTPS Web Server
Connectivity	Full inter-department ICMP verified

2. OBJECTIVES

- Design a scalable multi-department network topology using Cisco devices.
- Implement static IP addressing with a structured subnetting scheme across all departments.
- Configure a central router (Cisco 2811) to enable inter-department communication.
- Deploy a Server-PT with DNS service to resolve internal domain names.
- Host an HTTPS web server on the Server-PT accessible from all client PCs.
- Verify end-to-end connectivity using ICMP simulation in Packet Tracer.
- Document configurations for portfolio presentation and GitHub publishing.

3. NETWORK TOPOLOGY & ARCHITECTURE

The network follows a **star topology with a centralised routing model**. Router1 (Cisco 2811) acts as the core, with four 2960-24TT switches branching out — one per department. Each switch connects its local PC workstations and uplinks to Router1 on a dedicated subnet.

Architecture Breakdown:

Core Layer	Router1 (2811) — handles all inter-subnet routing via directly connected interfaces.
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Access Layer	Switch6, Switch7, Switch8, Switch9 (2960-24TT) — connect end devices per department.
Server Zone	Switch6 hosts Server-PT (192.168.1.100) alongside the management subnet.
End Points	7 PCs distributed across 3 non-management subnets (2 or 3 per department).

Logical Topology Diagram

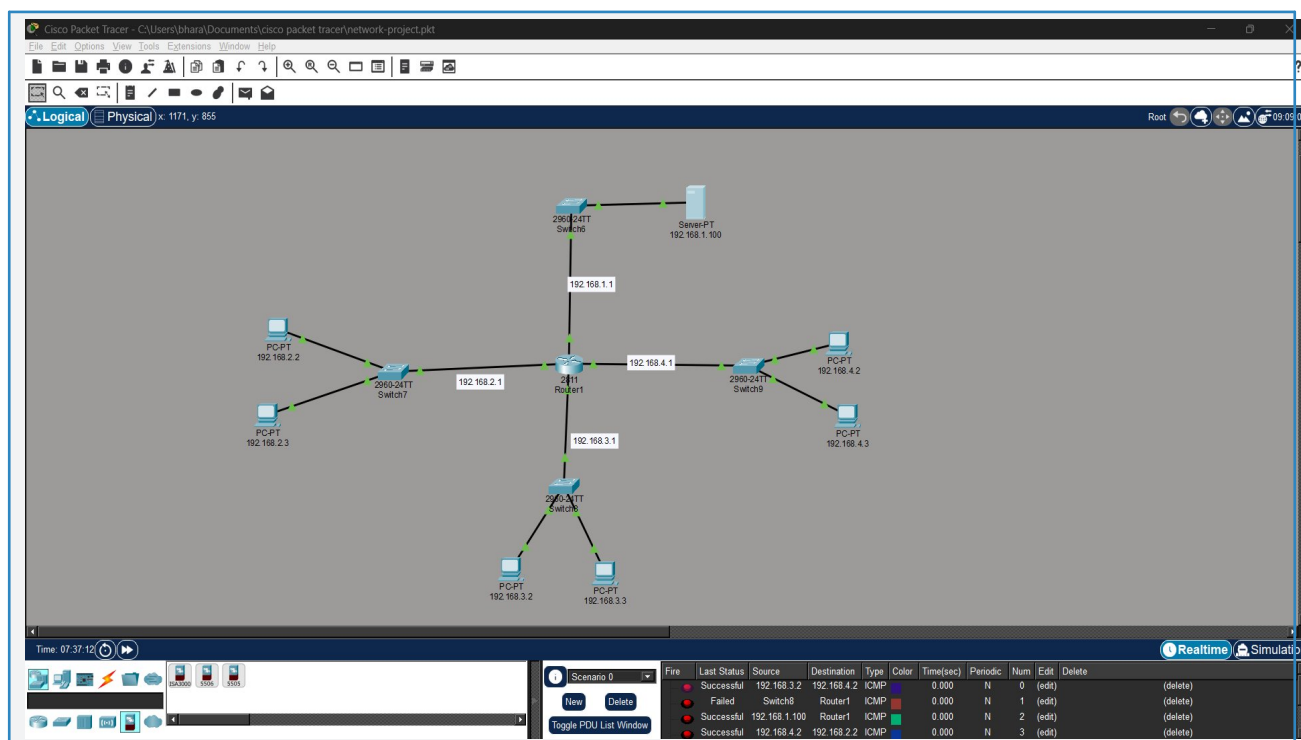


Figure 2 — Cisco Packet Tracer Logical View with PDU Simulation Results

4. IP ADDRESSING SCHEME

A Class C private address space (192.168.x.x/24) is used with four /24 subnets — one per department. Each subnet provides up to 254 usable host addresses, suitable for a small office segment.

Device	Interface / NIC	IP Address	Subnet Mask	Role
Router1	Fa0/0	192.168.1.1	255.255.255.0	Dept-1 / Mgmt GW
Router1	Fa0/1	192.168.2.1	255.255.255.0	Dept-2 GW
Router1	Fa0/2	192.168.3.1	255.255.255.0	Dept-3 GW
Router1	Fa0/3	192.168.4.1	255.255.255.0	Dept-4 GW
Server-PT	NIC	192.168.1.100	255.255.255.0	DNS + HTTPS Server
PC (Dept-2)	NIC	192.168.2.2	255.255.255.0	Workstation

Device	Interface / NIC	IP Address	Subnet Mask	Role
PC (Dept-2)	NIC	192.168.2.3	255.255.255.0	Workstation
PC (Dept-3)	NIC	192.168.3.2	255.255.255.0	Workstation
PC (Dept-3)	NIC	192.168.3.3	255.255.255.0	Workstation
PC (Dept-4)	NIC	192.168.4.2	255.255.255.0	Workstation
PC (Dept-4)	NIC	192.168.4.3	255.255.255.0	Workstation

5. DEVICE CONFIGURATION DETAILS

5.1 Router1 (Cisco 2811)

Router1 is the backbone of the network. Four FastEthernet interfaces are configured with gateway IPs for each subnet. Static routing is not required since all four subnets are directly connected to the router's interfaces — the router automatically populates its routing table with connected routes.

```
Router> enable
Router# configure terminal
Router(config)# hostname Router1

Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
Router1(config-if)# no shutdown

Router1(config)# interface FastEthernet0/1
Router1(config-if)# ip address 192.168.2.1 255.255.255.0
Router1(config-if)# no shutdown

Router1(config)# interface FastEthernet0/2
Router1(config-if)# ip address 192.168.3.1 255.255.255.0
Router1(config-if)# no shutdown

Router1(config)# interface FastEthernet0/3
Router1(config-if)# ip address 192.168.4.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# end
Router1# write memory
```

5.2 Switches (Cisco 2960-24TT)

All four switches (Switch6, Switch7, Switch8, Switch9) operate at Layer 2 in access mode. No VLAN trunk configuration is required in this flat-switched design — each switch simply forwards frames between its connected PCs and the router uplink. Default VLAN 1 is used across all switches.

Switch6 — Subnet: 192.168.1.x | Connected: Server-PT (192.168.1.100) — DNS + HTTPS

Switch7 — Subnet: 192.168.2.x | Connected: PC 192.168.2.2, PC 192.168.2.3

Switch8 — Subnet: 192.168.3.x | Connected: PC 192.168.3.2, PC 192.168.3.3

Switch9 — Subnet: 192.168.4.x | Connected: PC 192.168.4.2, PC 192.168.4.3

5.3 PC Workstations

Each PC-PT is statically configured with its IP address, subnet mask (255.255.255.0), and default gateway pointing to the Router1 interface on its subnet. DNS server is set to 192.168.1.100 (Server-PT) on all PCs to enable domain name resolution.

6. DNS & HTTPS SERVER SETUP

6.1 DNS Configuration

The Server-PT at 192.168.1.100 runs Packet Tracer's built-in DNS service. An A record maps a domain name (e.g., *intranet.company.local*) to the server's own IP address (192.168.1.100). All PCs across all departments are configured with 192.168.1.100 as their DNS server, allowing domain-name based access to the web server.

Step 1: Open Server-PT → Services tab → DNS

Step 2: Enable DNS Service (toggle ON)

Step 3: Add A Record: Name = "intranet.company.local" → Address = 192.168.1.100

Step 4: Save / Apply the record

Step 5: On each PC: Desktop → IP Configuration → DNS Server = 192.168.1.100

6.2 HTTPS Web Server Configuration

Server-PT's built-in HTTP/HTTPS service is enabled to host the internal company webpage. Both HTTP (port 80) and HTTPS (port 443) are activated. A basic HTML page is configured as the index, representing the company intranet portal.

Step 1: Open Server-PT → Services tab → HTTP

Step 2: Enable HTTP Service (ON) and HTTPS Service (ON)

Step 3: Edit index.html with a custom intranet homepage content

Step 4: Save the file

Step 5: From any PC: Desktop → Web Browser → type `https://192.168.1.100` OR `https://intranet.company.local`

Step 6: Verify the webpage loads successfully

✓ **Result:** All 7 PCs from departments 2, 3, and 4 can successfully browse to <https://intranet.company.local> — routed through Router1 across subnet boundaries.

7. INTER-DEPARTMENT CONNECTIVITY

All departments are connected through Router1, which maintains directly connected routes for all four subnets. Traffic between departments is routed at Layer 3. The PDU simulation panel in the screenshot confirms successful ICMP exchanges between non-adjacent subnets.

Source	Destination	Protocol	Result	Path
192.168.3.2	192.168.4.2	ICMP	✓ Successful	Dept3→Router1→Dept4
192.168.1.100	192.168.2.2	ICMP	✓ Successful	Server→Router1→Dept2
192.168.4.2	192.168.2.2	ICMP	✓ Successful	Dept4→Router1→Dept2
Switch8	Router1	ICMP	✗ Failed*	L2 switch ping to L3 (expected)

** Switch-to-router ICMP failure is expected — Layer 2 switches do not have IP routing capabilities by default in this configuration and do not respond to ICMP echo requests from other subnets.*

8. SECURITY CONSIDERATIONS

While this project focuses on connectivity and services, several security best practices are applicable and should be considered for production-grade deployment:

HTTPS over HTTP	HTTPS (port 443) is enabled on the server instead of plain HTTP, ensuring encrypted web traffic between clients and the server — mitigating eavesdropping.
Network Segmentation	Separating departments into distinct subnets limits broadcast domains and provides a foundation for implementing Access Control Lists (ACLs) to restrict inter-department traffic.
ACL Hardening (Recommended)	Router1 can be further hardened using extended ACLs to allow only specific traffic (e.g., allow HTTPS to server, deny all other cross-department traffic).
SSH over Telnet (Recommended)	Cisco IOS supports SSH for secure remote management. Telnet should be disabled on all devices and SSH configured with RSA keys for encrypted CLI access.
DNS Security	In production, DNSSEC should be used to prevent DNS spoofing. Internal DNS servers should not be exposed externally.
Physical Security	All network devices (router, switches) should be secured in locked cabinets with console port passwords and enable secret configured.

9. TESTING & VERIFICATION

Verification was performed using Packet Tracer's built-in Simulation Mode and PDU (Protocol Data Unit) testing. The PDU list panel visible in the project screenshot confirms multiple successful ICMP transactions across subnets.

Test Case	Method	Expected	Actual Result
PC Dept-2 → Server HTTPS	Web Browser	Page loads	✓ Pass
PC Dept-3 → Server DNS	nslookup / Browser	IP resolved	✓ Pass
PC Dept-4 → PC Dept-2 ping	ICMP PDU	Reply received	✓ Pass
PC Dept-3 → PC Dept-4 ping	ICMP PDU	Reply received	✓ Pass
Router1 routing table	show ip route	4 connected routes	✓ Pass
All PCs → Server ping	ICMP PDU	All replies	✓ Pass

10. SKILLS DEMONSTRATED

Network Design	Designed a multi-subnet enterprise topology from scratch.
IP Subnetting	Applied Class C /24 subnetting across 4 departments.
Cisco IOS	Configured interfaces, hostnames, and routing on Cisco 2811.
Inter-VLAN Routing	Enabled cross-subnet communication through router interfaces.
DNS Configuration	Set up internal DNS with A record resolution on Server-PT.
HTTPS Services	Deployed an HTTPS web server accessible enterprise-wide.
Network Verification	Used PDU simulation and ICMP to verify end-to-end connectivity.
Documentation	Produced structured technical documentation for GitHub portfolio.
Security Awareness	Identified applicable hardening measures (ACL, SSH, HTTPS).

11. CONCLUSION

This project successfully demonstrates the design and deployment of a small enterprise network simulation using Cisco Packet Tracer. A central router manages inter-department routing across four distinct subnets, while a dedicated server provides DNS resolution and HTTPS web services to all connected workstations.

The verified connectivity across all departments, combined with the DNS and HTTPS service deployment, mirrors real-world network environments encountered in corporate IT and SOC settings. This project serves as a practical demonstration of foundational networking skills essential for roles in network administration, cybersecurity analysis, and SOC operations.

Future enhancements could include: VLAN segmentation with 802.1Q trunking, extended ACLs for traffic filtering, SSH hardening on all devices, OSPF dynamic routing, and a DMZ zone for the server.

12. REFERENCES & RESOURCES

- Cisco Networking Academy — Cisco Packet Tracer (netacad.com)
- Cisco IOS Configuration Guide — 2800 Series Routers
- RFC 1918 — Address Allocation for Private Internets
- RFC 2535 — Domain Name System Security Extensions (DNSSEC)
- OWASP — Transport Layer Security Cheat Sheet
- CompTIA Network+ Study Guide — Mike Meyers
- TryHackMe — Pre-Security & Networking Fundamentals Path (tryhackme.com)
- Hack The Box Academy — Network Fundamentals Module (hackthebox.com)

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